

5.2

skladni' relacny pro $\lambda_{1,2} = 2$

$$v_1 = (1, 1, 0)^T$$

$$v_2 = (0, 1, 1)^T$$

$$\lambda_3 = 3$$

$$v_3 = (1, 1, 1)^T$$

$$\text{uplny' } x - y + z = 0$$

$$B = (v_1, v_2, v_3)$$

matice v bazi B: $(v_1, v_2, v_3) =: P$

$$D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$P = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$\det P = 1 \Rightarrow P^{-1} = \begin{pmatrix} 0 & 1 & -1 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix}$$

$$A = P D P^{-1} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 0 & 1 & -1 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix} = \dots = \begin{pmatrix} 3 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 3 \end{pmatrix}$$

$$A \vec{x} = P D (P^{-1} \vec{x})$$

$$[id]_K^B [f]_B^B [id]_B^K$$